

CPRD and HES Data Cleaning Steps and Assumptions

Step in data cleaning process	Cleaning step or assumption	Additional notes
Step 1: Patients	All cancer patients must have a non-cancer matched control	All non-cancer patients must also have a matched cancer patient All patients must have a patient ID number
Step 2: Cancer files	All cancer patients have exactly one cancer record	All cancer patients must have a cancer diagnosis. All cancer records must have a cancer code. First cancer diagnosis record selected. For those with multiple same day diagnoses, record with highest cancer stage selected, otherwise one record selected at random.
	Patients diagnosed with 2WW referrals, routine referrals, emergency, inpatient or outpatient appointments only	
	All patients are aged over 40 at diagnosis	For both case and control patients
	All dates of death must be after the patient's diagnosis date	For both case and control – control dates of death must be after the matched case diagnosis date
	Patients' GP registration date must be more than 3 years before diagnosis date and not after	For both case and control patients – control GP registration date must be before matched case diagnosis date
	Cancer stages are simplified 0,1, A -> 1 2 -> 2 3, B -> 3 4, C -> 4	Those listed as '?', ' ', U, X, 6 or NA are relabelled as NA. Stages 1 and 2 are 'early stage'. Stages 3 and 4 are 'late stage'
Step 3a: A/D from observations	Observations for pre-existing anxiety or depression must occur 1-2 years before the patient's cancer diagnosis date	All observations must occur before the patient's diagnosis date.

		Anxiety and depression observations are identified using code lists
	Anxiety and or depression patients must have at least one related observation 1-2 years prior to cancer diagnosis	
Step 3b: A/D from drugs issued	Drugs issued for pre-existing anxiety or depression must occur 1-2 years before the patient's cancer diagnosis date	All drugs must be issued before the patient's diagnosis date. Anxiety and depression related drugs are identified using code lists
	Anxiety and or depression patients must have at least four related drugs issued 1-2 years prior to cancer diagnosis	
Step 4: Consultation files	All GP consultations must occur within one year prior to the patient's cancer diagnosis	
	All consultations must be classed as face-to-face with a GP	Face-to-face consultations with a GP are identified using code lists
	Cancer patients must have at least one GP consultation prior to diagnosis	For multiple consultation records on the same day, only the first appointment is selected
Step 5: Symptoms from observations	Symptoms of cancer must be recorded within 1 year of patient's diagnosis date	Control patients must have symptoms in the same window as case patient
	All cancer related observations must either be a cancer symptom or a referral	Cancer related symptoms are identified using code lists
Step 6: CPRD cancer referrals from symptoms in observation files	Referrals must be within 1 year of patient's cancer diagnosis date	Control patients must have referrals in the same window as case patient For multiple referrals on the same day, only first referral is selected
	Referrals with 2WW (USC) priority are immediately taken to be cancer related	
	Referrals with non-2WW priority are only cancer related if there is a cancer symptom observation and a	Window is taken to be 7 or 14 days prior to referral date

	related GP consultation within a window of the referral date	
Step 7: HES outpatient appointments	All appointments must come from a GP referral	
	All appointments must be requested from a GP within 1 year before patient's cancer diagnosis	Control patients must have appointments in the same window as case patient
	All appointments must be attended or not attended without reason	Appointments that are cancelled or unknown attendance are removed
	All follow-up appointments must be preceded by a first appointment, and the requested date must be on or after the requested date of the corresponding first appointment	
Step 8: Cancer referrals from HES outpatient appointments	Referral appointments with 2WW (USC) priority are immediately taken to be cancer related	
	Referral appointments with non-2WW priority are only cancer related if there is a cancer symptom observation, a related GP consultation within a window of the referral date, and the cancer symptom observation corresponds with the referral appointment treatment specialist	Window is taken to be 7 or 14 days prior to referral date Cancer related symptoms corresponding to treatment specialist taken from clinical opinion
Step 9: Further data cleaning	Multiple referral and referral appointments that overlap, where the referral requested date is the same are related to the same patient issue	Take the first appointment that occurs, then only keep the referrals that match to the appointment referral date If appointments have different priorities, select highest priority (2WW/USC)
	Multiple first consecutive referral appointments with the same referral requested date are related to the same patient issue	Take the first appointment that occurs, and define subsequent appointments to be follow-ups If appointments have different priorities, select highest priority (2WW/USC)
	All referral appointments must have only one referral and the requested date must match the referral date with the same priority	If appointments and referrals have different priorities, select highest priority (2WW/USC)
	All referrals must have a date that precedes the corresponding referral appointment	Sample a new referral date from known wait times if incorrect

	A referral appointment must exist and occur before either a 2WW/USC or non-2WW diagnosis	Sample a new referral appointment date (and follow-ups) from known wait times if incorrect/missing
	The priority of the route to diagnosis must match the preceding referral appointment	If appointments and diagnosis have different priorities, select highest priority (2WW/USC)
	Dates of referrals, appointments and diagnosis must come before patient's date of death	Sample new referral, referral appointments and/or diagnosis dates from known wait times if incorrect/missing
	Previous GP consultation date must be the same as the referral date	
	There is no more than 90 days between a 2WW/USC referral appointment and a 2WW diagnosis	Cap the time between appointment and diagnosis
	All patient pathways must begin with a GP consultation	Remove all events that occur before the first GP consultation
Step 10: Modelling choices	All GP consultations between referrals and referral appointments are ignored	
	All GP consultations between referral appointments and a related 2WW or non-2WW diagnosis are ignored	
	Appointments recorded as did not attend (DNA) are not modelled directly for simplicity	DNA events are ignored causing long wait times between GP referrals and referral appointments
	The times of events that should occur on the same day are shifted by short periods	Previous events occur ~1 hour earlier Required for multi-state modelling only
	The number of GP appointments is not modelled directly	The time spent at in the GP consultation is modelled instead Required for multi-state modelling only
	Patients are not permitted to transition from a referral directly to a GP consultation	Choice due to minimal data
	Patients are not permitted to transition from a referral or referral appointment directly to no diagnosis	Choice due to minimal data